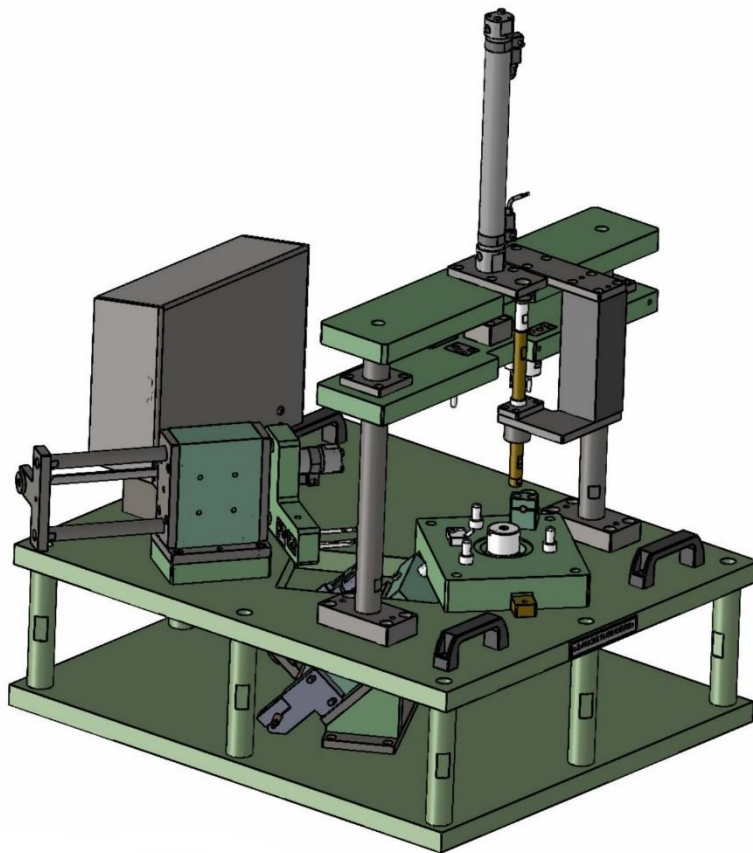


OPERATION AND MAINTENANCE MANUAL

2.4 FIXTURE



December, 2024

MINDA CORPORATION LTD

DOCUMENT INFORMATION

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1.GENERAL INFORMATION

- ❖ This manual provides comprehensive guidelines for the operation, maintenance, and troubleshooting of ECM (Electro chemical Machining) fixtures.
- ❖ It is essential for operators and maintenance personnel to thoroughly understand and follow the instructions in this manual to ensure safe and efficient operation of the fixtures.

2.GENERAL SAFETY WARNINGS

- ❖ Personal Protective Equipment (PPE): Always wear appropriate PPE, including safety goggles, gloves, and protective clothing, when operating ECM fixtures.
- ❖ Electrical Safety: Ensure that the power supply is turned off before performing any maintenance. Use insulated tools and follow lockout/tag-out procedures.
- ❖ Chemical Handling: Handle electrolytes and other chemicals with care. Follow the Material Safety Data Sheets (MSDS) for proper storage, handling, and disposal.
- ❖ Emergency Procedures: Familiarize yourself with emergency shut-off procedures, first aid, and fire safety protocols.

3.OVERVIEW OF ECM FIXTURES

- ❖ ECM fixtures are designed to hold and position workpieces during the electrochemical machining process.
- ❖ They play a critical role in ensuring precision and consistency in machining operations.
- ❖ The fixtures are typically made from non-corrosive materials and are equipped with features to facilitate easy workpiece mounting, alignment, and removal.

4. GENERAL OPERATION GUIDELINES

4.1. Pre-Operation Checklist

- ❖ Inspect the ECM fixture for any visible damage or wear.
- ❖ Ensure that all connections including electrical, pneumatic and electrolyte lines for any visible damaged are view.
- ❖ Verify that the workpiece is correctly positioned and securely clamped in the fixture.
- ❖ Check that the electrolyte flow system is functioning correctly and that the electrolyte level is sufficient.
- ❖ Ensure that the power supply and control systems are functioning properly.

4.2. Deburring Process

4.2.1 Cycle Start-Up

- ❖ Turn on the ECM machine and initialize the fixtures.
- ❖ Select the appropriate deburring parameters (Voltage, Current, Electrolyte Flow, Deburring Time) as specified for the workpiece.
- ❖ Begin the Deburring process by starting the auto cycle.

4.2.2 Deburring

- ❖ Monitor the deburring process closely, ensuring that the electrolyte flow is consistent and that there are no blockages.
- ❖ Observe the fixture and workpiece for any signs of abnormal wear, vibration, or overheating.
- ❖ Make adjustments to the process parameters if necessary to maintain the precise deburring.

4.2.3 Cycle End

- ❖ Upon completion of the deburring process, the autocycle end.
- ❖ Carefully remove the workpiece from the fixture, ensuring that no damage occurs during removal.
- ❖ Wash with DM water after cycle complete.

5. MAINTENANCE PROCEDURES

5.1 Regular Maintenance

5.1.1 Daily

- ❖ Clean the fixture and surrounding area with DM water to remove any residual electrolyte or debris.
- ❖ Inspect the fixture for signs of corrosion, wear, or damage. Replace any worn or damaged components immediately.

5.1.2 Weekly

- ❖ Check the alignment and condition of clamping mechanisms. Lubricate moving parts as required.
- ❖ Inspect electrical connections for signs of wear or corrosion. Tighten or replace connectors as needed.

5.1.3 Monthly

- ❖ Perform a thorough inspection of the entire fixture, including all electrical and mechanical components.
- ❖ Test the electrolyte flow system for proper operation. Replace filters and clean lines as necessary.

5.2 Preventive Maintenance

- ❖ Refer the Preventive maintenance check list given by Vamtec Machine.
- ❖ Schedule periodic checks of the ECM machine and fixtures by qualified personnel.
- ❖ Replace worn components according to the manufacturer's recommendations.
- ❖ Keep a detailed maintenance log to track the condition of the fixture and any repairs or replacements performed.

6. TROUBLESHOOTING

6.1 Fixture Not Holding Workpiece

- ❖ Check for wear or damage to clamping mechanisms.
- ❖ Ensure the workpiece is properly aligned and seated in the fixture.

6.2 Electrolyte Flow Issues

- ❖ Inspect for blockages in the electrolyte lines and electrolyte fittings.
- ❖ Check the electrolyte pump operation.

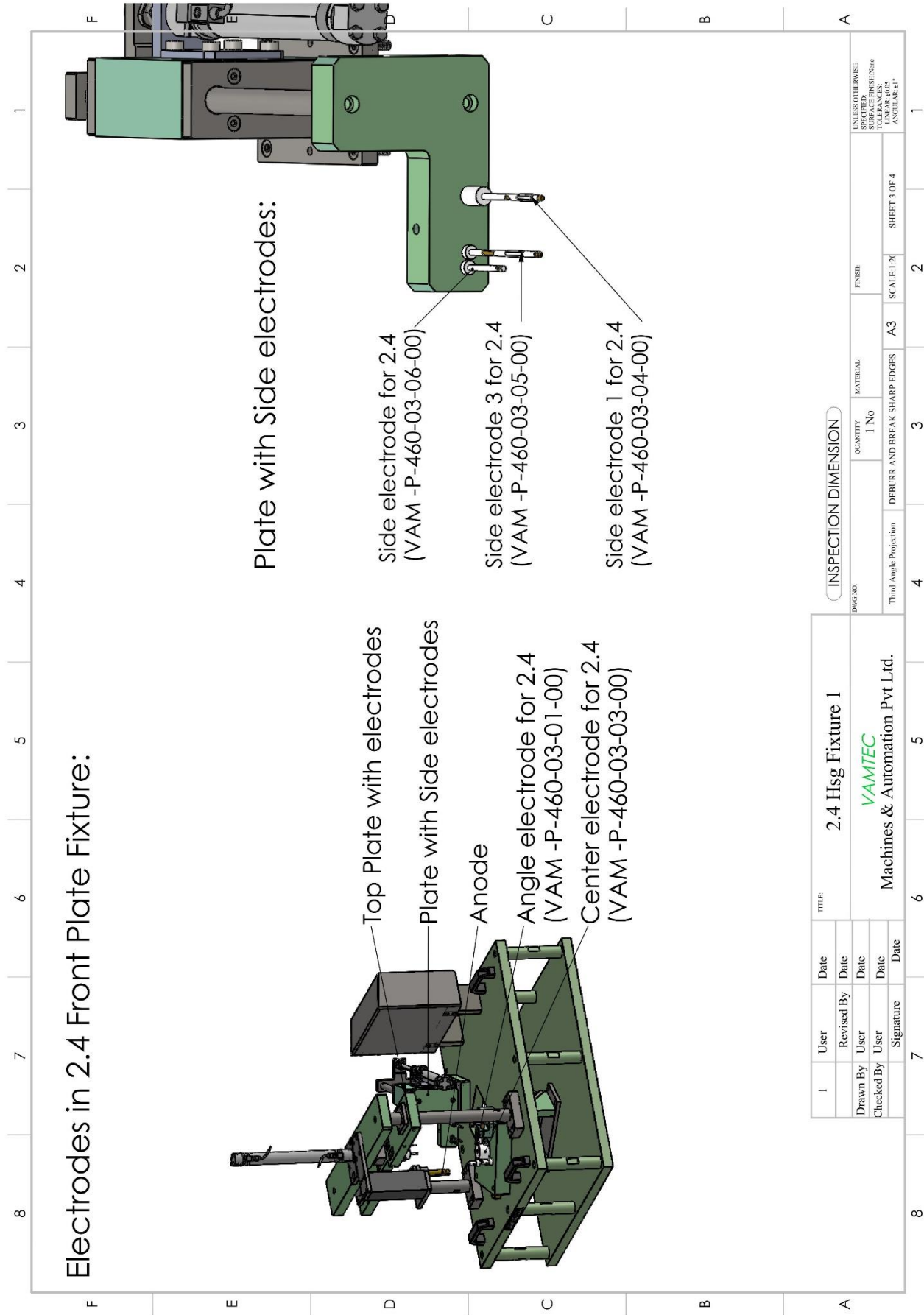
6.3 Short Circuit Issues

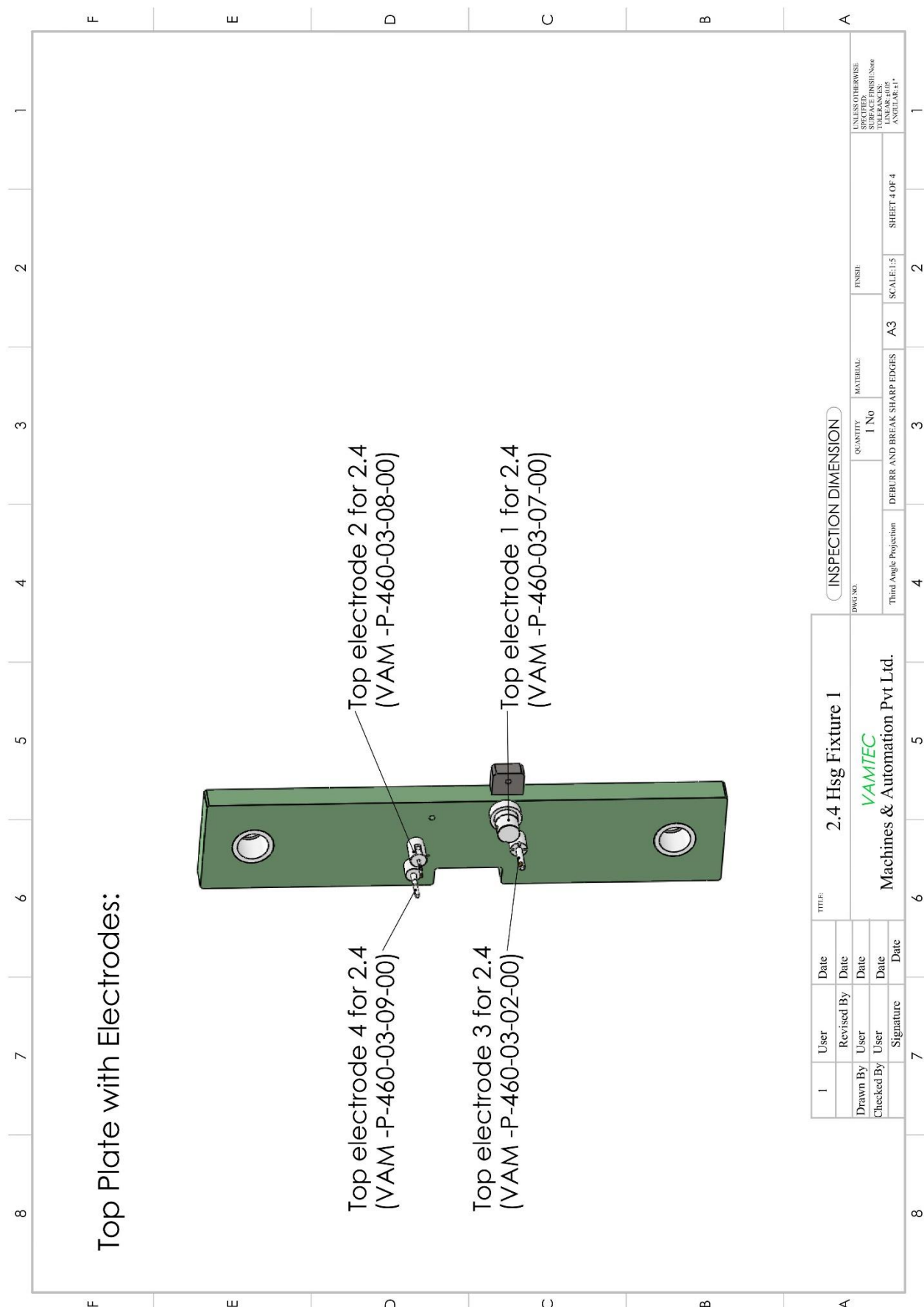
- ❖ Check the burr in the workpiece.
- ❖ Check the alignment of the workpiece and electrodes

TITLE:		INSPECTION DIMENSION	
Machines & Automation Pvt Ltd.	DWG NO.	QUANTITY	FINISH
		Nos	
	Third Angle Projection	DEBURR AND BREAK SHARP EDGES	SCALE: 1:20
			SHEET 1 OF 4

1	2	3	4	5	6
1	User	Date	1	User	Date
	Revised By	Date		Revised By	Date
	Drawn By	User		Drawn By	User
	Checked By	User		Checked By	User
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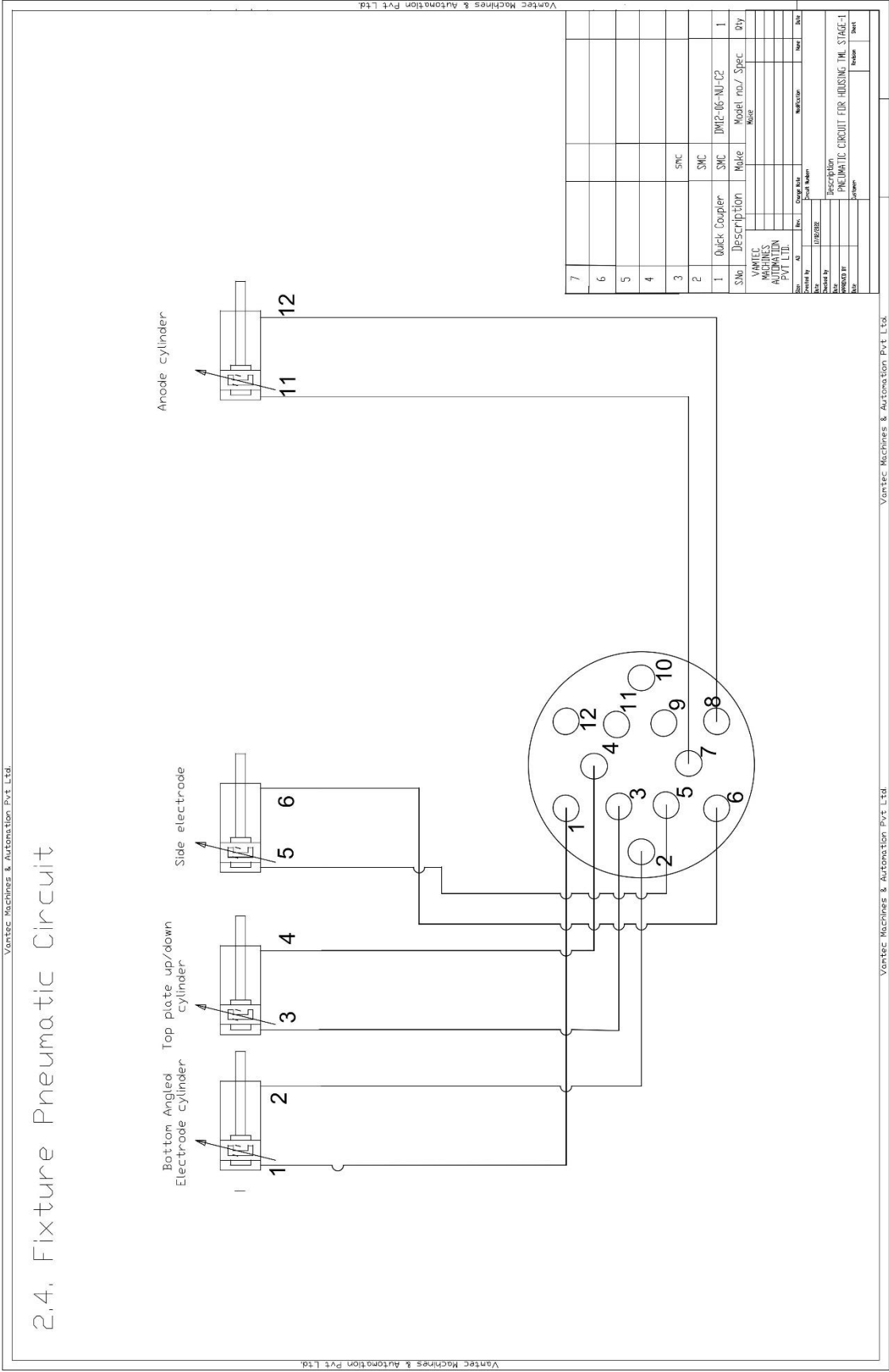


1		User	Date	TITLE: 2.4 Hsg Fixture 1	
Drawn By		Revised By	Date	Machines & Automation Pvt Ltd. VAMTEC	
Checked By		User	Date		
		Signature	Date		

INSPECTION DIMENSION

Third Angle Projection	DEBURR AND BREAK SHARP EDGES	MATERIAL:	FINISH:	QUANTITY 1 No	SCALE:1:5	SHEET 4 OF 4	UNLESS OTHERWISE SPECIFIED: SURFACE FINISH:None TOLERANCES: ANGULAR:1°
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8. PNEUMATIC CONNECTION DETAILS



9. WEAR PARTS LIST

S.NO	Part Name	Drawing number
1	Connector for anode 2.4	VAM -P-460-02-06-00
2	Anode guide bush for 2.4	VAM -P-460-02-05-00
3	Changeable anode for 2.4	VAM -P-460-02-08-00
4	Locating pin for 2.4	VAM -P-460-02-27-00
5	Peek bush for Anode (460)	VAM -P-460-02-15-00
6	Peek Bush in electrode mounting block	VAM-P-427-03-09 00